

14.7 Maximum and Minimum Values

1 Warm Up

Problem: Solve the system of equations.

$$\begin{cases} 3x + 4y - 5 = 2x + 1 \\ 4y - x = y + 2x + 1 \end{cases}$$

Problem Find all relative max/min if $f(x) = x^5/5 + 5x^4/4 + 2x^3$

2 Motivation

In 2D whenever our derivative is equal to zero, we have a maximum or minimum value i.e. a peak or a valley. In 3D what does it mean for our derivative to be zero? Do we have a notion of a local maximum or minimum? In this section we utilize our knowledge of partial derivative to find local minimums and maximums and even more!

3 Definitions & Theorems

1. In calc 1 we learn that **relative** min/max are given by critical points are when the derivative is zero or it is undefined. Occasionally they could also be inflection points when I change concavity. In calc 3 *both* f_x and f_y must equal zero at the same point in order to have a critical value. Or similarly a point which is undefined or a saddle point (changes concavity). *Sketch a picture to see why you don't just need f_x or f_y .*
 - Both f_x and f_y are Increasing $\rightarrow$ Decreasing: Relative Maximum
 - Both f_x and f_y are Decreasing $\rightarrow$ Increasing: Relative Minimum
 - One changes from Increasing $\rightarrow$ Decreasing and the other from Decreasing $\rightarrow$ Increasing: Saddle Point
2. If first derivatives tell us about increasing and decreasing, then second derivatives tell us about concavity and will give us a way to determine if we have a maximum/minimum/saddle. This is known as the second derivative test and requires to compute the following.

$$D(x, y) = f_{xx}(x, y)f_{yy}(x, y) - (f_{xy}(x, y))^2.$$

The test says the following for the 4 possible outcomes.

- $D(a, b) > 0, f_{xx}(a, b) > 0 \implies (a, b)$ relative minimum
 - $D(a, b) > 0, f_{xx}(a, b) < 0 \implies (a, b)$ relative maximum
 - $D(a, b) < 0 \implies (a, b)$ is a saddle point
 - $D(a, b) = 0 \implies$ Inconclusive
3. If we wish to find absolute maximums or minimums over some closed region then not only do we need to check critical values, we also need to check the boundary of our region. If our boundary of the region is created by lines, we can use these lines and substitute them into our original surface to get a curve! We then can use calc 1 knowledge of absolute max/min values on a curve.
4. Here is some helpful tips for finding absolutes.
- Draw the region
 - Label the endpoint of the region
 - Label your lines.

4 Examples

Problem 1. *Find and classify all critical values of $f(x, y) = (y - 2)x^2 - y^2$.*

Problem 2. *Find and classify all critical values of $f(x, y) = 7x - 8y + 2xy - x^2 + y^3$.*

Problem 3. Suppose $f_x = 3x^2 - 3y$ and $f_y = -3x + 3y^2$. Find and classify the critical values.

Problem 4. Suppose $f_x = 2x - 6 - \sqrt{y}$ and $f_y = -\frac{x}{2\sqrt{y}} + 1$. Find and classify the critical values.

Problem 5. Suppose $f_x = 3y - 2xy - y^2$ and $f_y = 3x - x^2 - 2xy$. Find and classify the critical values.

Problem 6. Find and classify all critical points for the function $f(x, y) = x^2 + 5y^2 + x^2y + 2y^3$.

Problem 7. Find the absolute maximum/minimum and where they occur on the surface $f(x, y) = 3x^2 + 2xy + y^2$ bounded by the triangle with end points $(-2, -1), (1, -1), (1, 2)$.

Problem 8. Find the absolute maximum/minimum and where they occur on the surface $f(x, y) = x^2 + xy + y^2$ on the region $R = \{-2 \leq x \leq 2, -1 \leq y \leq 1\}$

5 Scratch Work